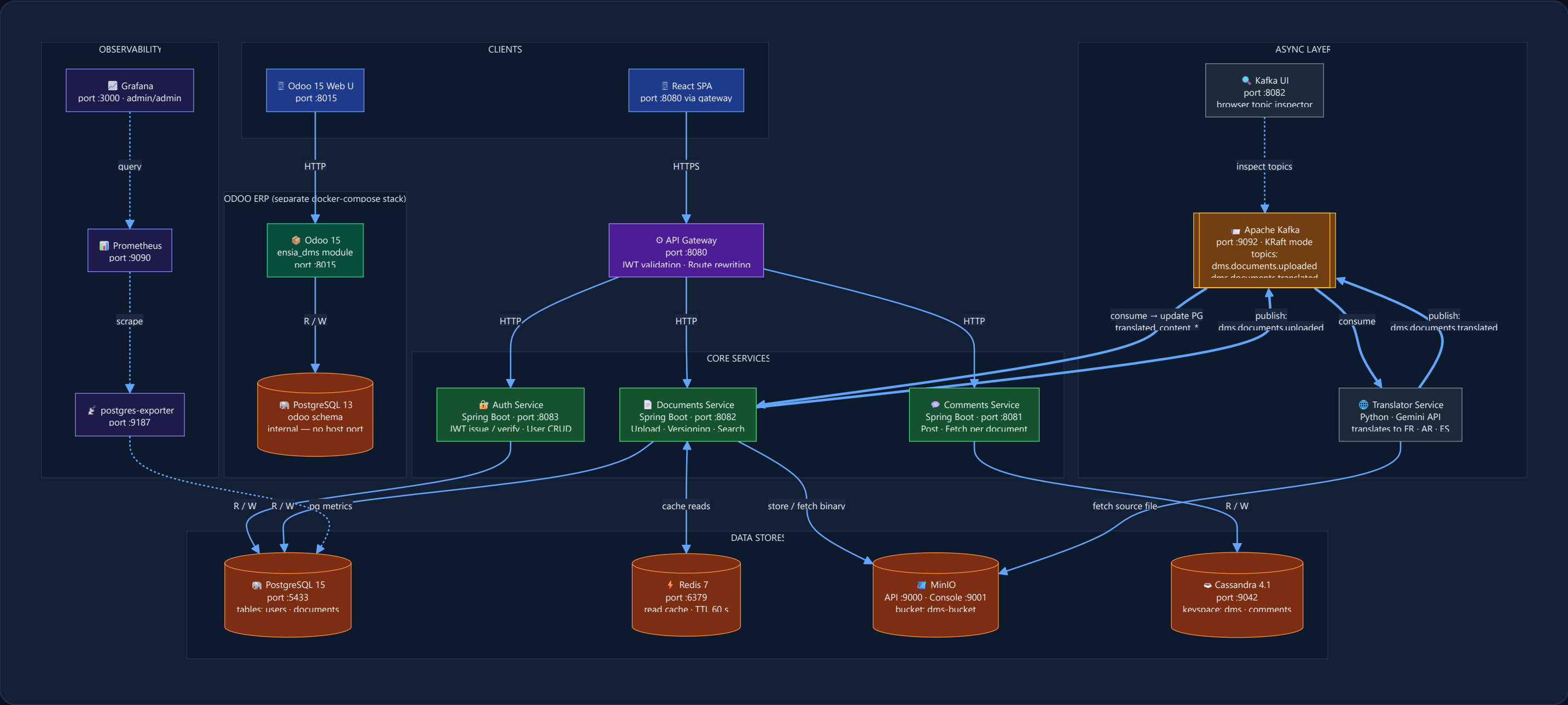
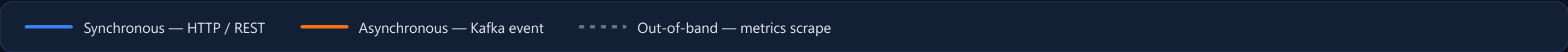


ENSIA DMS — System Architecture

Full-stack Document Management System · Microservices · Event-driven · 4th-Year Enterprise Computing Lab



FLOW SUMMARY			PORT REFERENCE		
FLOW	TYPE	PATH (ABBREVIATED)	SERVICE	HOST PORT(S)	NOTE
Login / JWT issue	Sync	Browser → GW → Auth → PG	API Gateway	8080	Single public entry point

Flow	Type	Path (Abbreviated)
Document upload	Sync	Browser → GW → Documents → PG + MinIO
Document read (cache hit)	Sync	Browser → GW → Documents → Redis
Document read (cold)	Sync	Browser → GW → Documents → Redis miss → PG
Comment post / fetch	Sync	Browser → GW → Comments → Cassandra
Translation trigger	Async	Documents ==> Kafka <code>dms.documents.uploaded</code>
Translation execution	Async	Kafka ==> Translator → MinIO + Gemini API
Translation persistence	Async	Translator ==> Kafka <code>dms.documents.translated</code> ==> Documents → PG
ERP document management	Sync	Odoo UI → Odoo 15 (<code>ensia_dms</code>) → PG (odoo schema)
Metrics collection	Out-of-band	Prometheus → postgres-exporter → PG → Grafana

Service	Host Port(s)	Note
Auth Service	8083	Also internal to GW
Documents Service	8082	Internal only
Comments Service	8081	Internal only
Odoo 15	8015	Separate compose stack
Kafka	9092 / 29092	External / internal broker
Kafka UI	8082	Dev tool
MinIO API	9000	S3-compatible
MinIO Console	9001	Web UI
PostgreSQL (DMS)	5433	Mapped from :5432
PostgreSQL (Odoo)	—	Internal only
Redis	6379	
Cassandra	9042	
Prometheus	9090	
Grafana	3000	admin / admin
postgres-exporter	9187	